

HACKING FOR DEFENSE MEMORANDUM

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FROM: Team Ad Astra - UC Berkeley Hacking for Defense, Spring 2026

SUBJECT: Project Executive Summary

- 1. Purpose:** This memorandum summarizes Ad Astra's findings while identifying logistics challenges for Space Force launch operations. In accordance with Executive Order 14369 on Ensuring American Space Superiority, this project aims to support the efficiency and sustainability of launch and exploration architectures.
- 2. Key Findings:** The team's discovery revealed that the reconciliation between divergent requirements, such as differences in payload guides, from a growing fleet of Launch Service Providers (LSPs) represents a risk in operational efficiency, including mission delays, rejected payloads, and compromised mission timelines. Space Logistics Officers currently lack a standardized interface to translate proprietary commercial standards (SpaceX, Blue Origin, etc.) into military compliance frameworks (MIL-STD-2073). They must parse and understand hundreds of pages of requirements for each LSP and its associated launch vehicle(s); yet, most logisticians, primarily from the Air Force, lack the experience and institutional knowledge due to their terrestrial logistics background. As the Space Force shifts toward a higher launch cadence with diverse commercial partners, this "translation gap" will only increase risks and pain points.
- 3. Solution:** An integration of a translation widget leveraging Machine Learning (ML) and Large Language Model (LLM) to increase efficiency while parsing through packaging requirements from LSPs. Ad Astra has developed a Minimum Viable Product (MVP), testing requirements from Lane 2 LSPs in an AWS Bedrock workshop to create a training tool. This MVP allows users to query the technical requirements of each LSP's documentation to determine mission requirements for any given launch. It provides logisticians who are unfamiliar with space operations with greater efficiency through a learn-by-doing approach, effectively helping them quickly build institutional knowledge while reducing risk and error rates.
- 4. Benefits to US Space Force and Commercial LSPs:** The translation widget delivers immediate value: (1) drastically reduces manual review time; (2) closes the MIL-STD/commercial LSP translation gap; (3) improves requirement parsing accuracy; and (4) reduces friction points while engaging with diverse LSPs.
- 5. Next Steps:** Ad Astra will migrate the MVP to a secure environment for live testing, integrate broader Lane 2 data to enhance accuracy, and evaluate funding opportunities. Additionally, data collected from the LLM may be used for further war gaming purposes and to better understand similar challenges the USSF may be facing.

It has been an honor to work on this project. Ad Astra remains committed to supporting the U.S. Space Force mission, as our nation advances into the next era of space.